



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
 - - Data collection | - Data wrangling | - EDA with Visualization | - EDA with SQL | - Interactive map with Folium | - Dashboard with Plotly Dash | - Predictive analysis (Classification)
- Summary of all results
 - - Exploratory Data Analysis results | - Interactive analytics demo in screenshots | - Predictive analysis results

Introduction

- Project background and context
- SpaceX is the most successful company of the commercial space age, making space travel affordable. Falcon 9 rocket launches cost \$62M vs competitors at \$165M+. Much of the savings is because SpaceX can reuse the first stage. Using public information and ML models, we predict if SpaceX will reuse the first stage.
- Questions to be answered
 - - How do payload mass, launch site, number of flights, and orbits affect first stage landing success?
 - - Does the rate of successful landings increase over the years?
 - - What is the best algorithm for binary classification in this case?

Section 1

Methodology

Methodology

Executive Summary

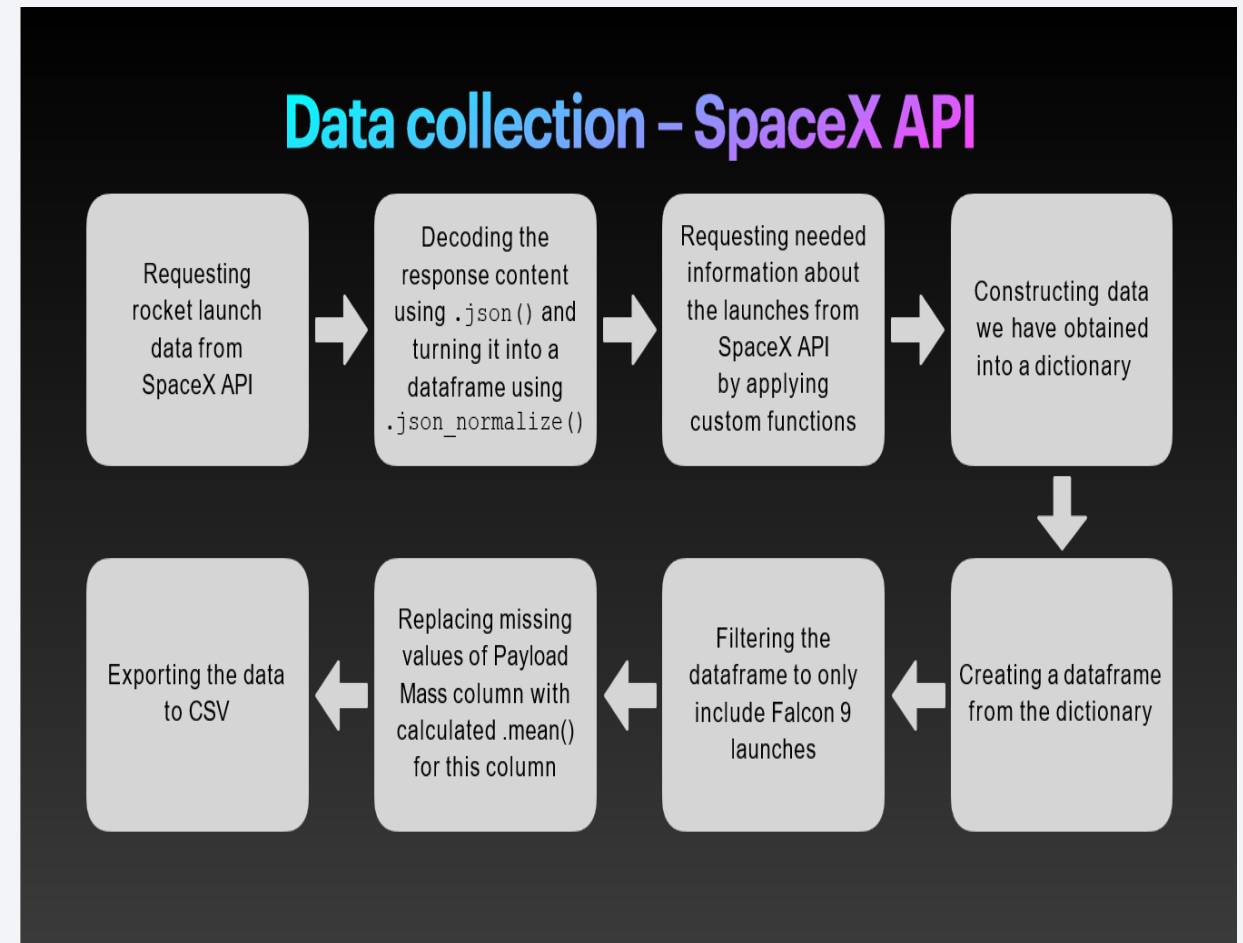
- Data collection methodology:
 - Using SpaceX REST API and Web Scraping from Wikipedia
- Performed data wrangling
 - Filtering, missing values handling, One Hot Encoding for binary classification
- Performed EDA using visualization and SQL
- Performed interactive visual analytics using Folium and Plotly Dash
- Performed predictive analysis using classification models
 - Building, tuning, and evaluation of classification models: LogReg, SVM, Decision Tree, KNN

Data Collection

- Describe how data sets were collected.
- You need to present your data collection process use key phrases and flowcharts

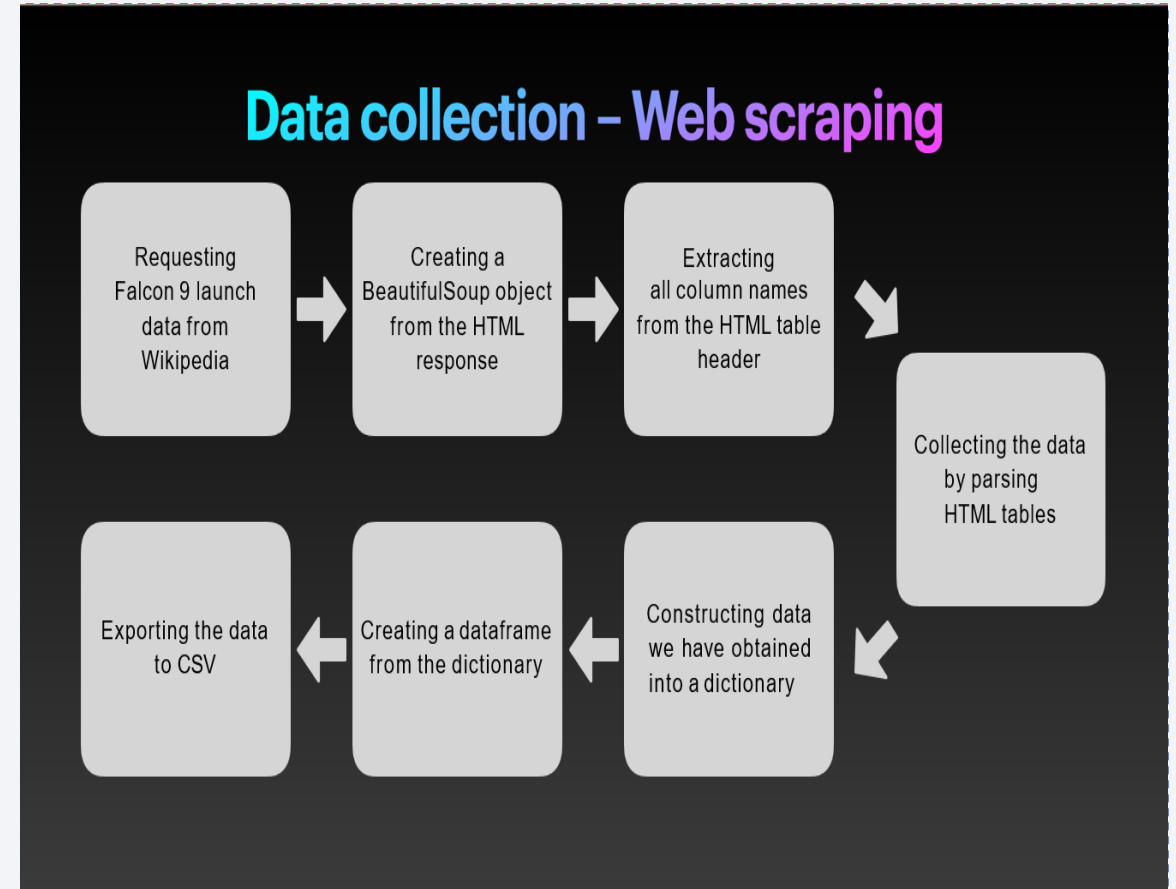
Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts
- Add the GitHub URL of the completed SpaceX API calls notebook (must include completed code cell and outcome cell), as an external reference and peer-review purpose



Data Collection - Scraping

- Present your web scraping process using key phrases and flowcharts
- Add the GitHub URL of the completed web scraping notebook, as an external reference and peer-review purpose



Data Wrangling

- Describe how data were processed
- You need to present your data wrangling process using key phrases and flowcharts
- Add the GitHub URL of your completed data wrangling related notebooks, as an external reference and peer-review purpose

EDA with Data Visualization

- Summarize what charts were plotted and why you used those charts
- Add the GitHub URL of your completed EDA with data visualization notebook, as an external reference and peer-review purpose

EDA with SQL

- Using bullet point format, summarize the SQL queries you performed
- Add the GitHub URL of your completed EDA with SQL notebook, as an external reference and peer-review purpose

Build an Interactive Map with Folium

- Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map
- Explain why you added those objects
- Add the GitHub URL of your completed interactive map with Folium map, as an external reference and peer-review purpose

Build a Dashboard with Plotly Dash

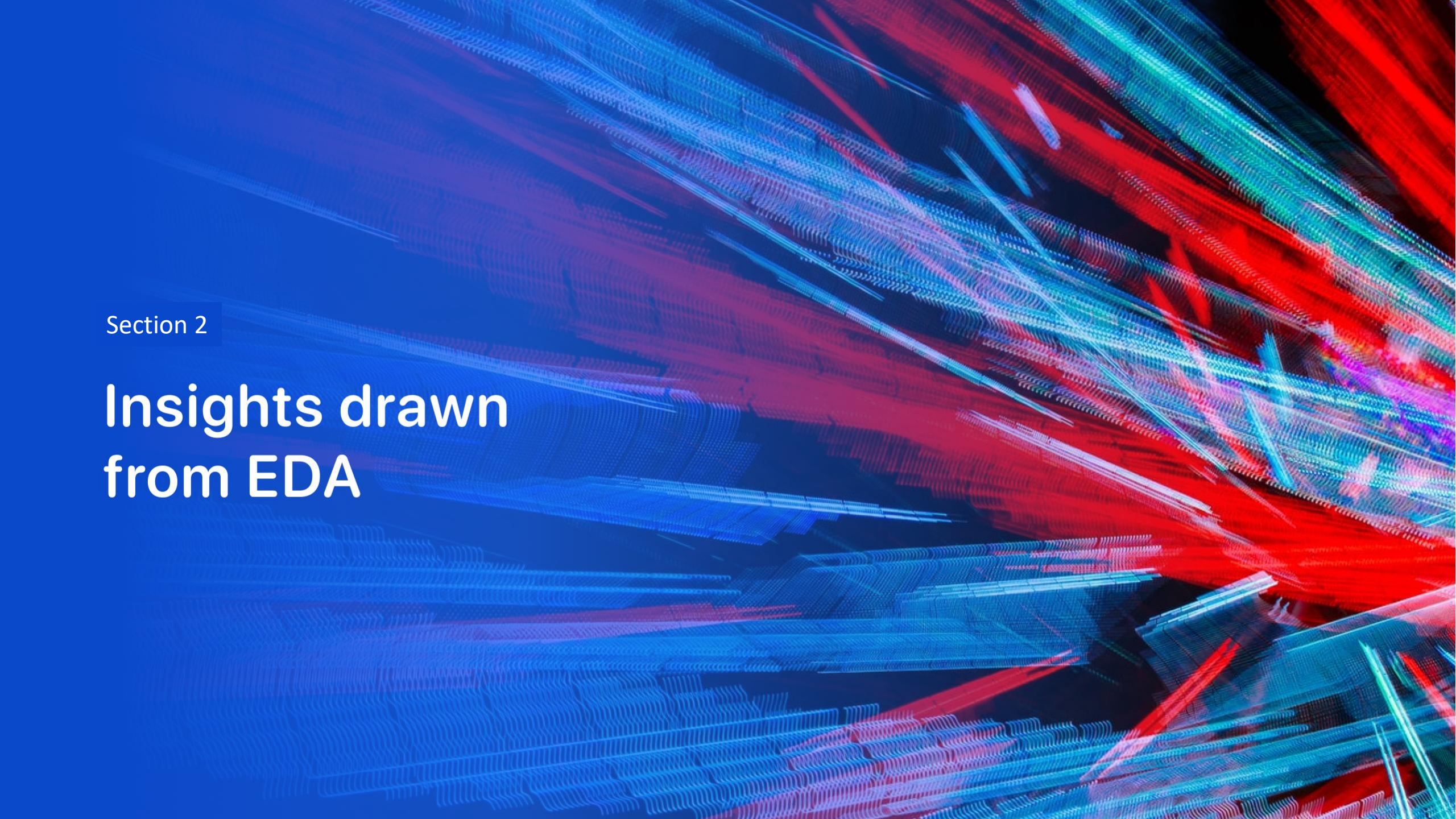
- Summarize what plots/graphs and interactions you have added to a dashboard
- Explain why you added those plots and interactions
- Add the GitHub URL of your completed Plotly Dash lab, as an external reference and peer-review purpose

Predictive Analysis (Classification)

- Summarize how you built, evaluated, improved, and found the best performing classification model
- You need present your model development process using key phrases and flowchart
- Add the GitHub URL of your completed predictive analysis lab, as an external reference and peer-review purpose

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

- Scatter plot: Flight Number vs. Launch Site - shows success improving with flight number.
- Earliest flights all failed; latest flights all succeeded. CCAFS SLC 40 has ~50% of all launches. VAFB SLC 4E and KSC LC 39A show higher success rates. Success rate increases with flight number.

Payload vs. Launch Site

- Scatter plot: Payload Mass vs. Launch Site - higher payload correlates with higher success.
- Higher payload mass correlates with higher success rate across all sites. Most launches with payload >7000 kg were successful. KSC LC 39A has 100% success rate even for payload under 5500 kg.

Success Rate vs. Orbit Type

- Show a bar chart: Success Rate per Orbit Type —
100% success: ES-L1, GEO, HEO, SSO | 0% success: SO
| 50-85%: GTO, ISS, LEO, MEO, PO
- 100% success: ES-L1, GEO, HEO, SSO. 0% success: SO.
50-85% success: GTO, ISS, LEO, MEO, PO.

Flight Number vs. Orbit Type

- Show a scatter plot of Flight Number vs. Orbit Type — LEO success correlates with more flights; no such trend in GTO orbit
- LEO orbit success is related to number of flights (more experience = more success). GTO orbit shows no such relationship with flight number.

Payload vs. Orbit Type

- Show a scatter plot of Payload Mass vs. Orbit Type
— heavy payloads hurt GTO outcomes but benefit LEO/ISS orbits
- Heavy payloads negatively influence GTO orbit outcomes but positively influence Polar LEO (ISS) orbit outcomes.

Launch Success Yearly Trend

- Show a line chart of Launch Success Yearly Trend — rose from 0% in 2010-2013 to ~90% by 2019-2020
- Success rate was 0% until 2013, then steadily increased each year through 2019-2020, reaching ~90%. Each new launch has a higher probability of success.

All Launch Site Names

- 4 unique launch sites identified:
- CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E

Launch Site Names Begin with 'CCA'

- 5 records where launch sites begin with 'CCA':
- All 5 records are from CCAFS LC-40, dates from 2010-06-04 to 2013-03-01, payloads 0-677 kg, all in LEO orbit for NASA (COTS/CRS).

Total Payload Mass

- Total payload mass by NASA (CRS) boosters:
- Total payload mass = 45,596 kg. All CRS missions launched from CCAFS LC-40 to ISS orbit.

Average Payload Mass by F9 v1.1

- Average payload mass by booster version F9 v1.1:
- Average payload mass = 2,534 kg for F9 v1.1 booster version.

First Successful Ground Landing Date

- First successful landing outcome on ground pad:
- Date: 2015-12-22. This was a historic milestone for SpaceX reusable rocket technology.

Successful Drone Ship Landing with Payload between 4000 and 6000

- Boosters with drone ship success and payload 4000-6000 kg:
- F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Total mission outcomes:
- Success: 99 missions | Failure (in flight): 1 | Success (payload status unclear):
 1. Overall mission success rate is extremely high at ~98%.

Boosters Carried Maximum Payload

- Booster versions carrying maximum payload mass:
- F9 B5 series: B1048.4, B1049.4, B1051.3, B1056.4, B1048.5, B1051.4, B1049.5, B1060.2, B1058.3, B1051.6, B1060.3, B1049.7. All are Block 5 boosters capable of carrying the heaviest payloads.

2015 Launch Records

- Failed drone ship landings in 2015:
- January 2015-01-10: F9 v1.1 B1012, CCAFS LC-40, Failure (drone ship)
- April 2015-04-14: F9 v1.1 B1015, CCAFS LC-40, Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Landing outcome counts 2010-06-04 to 2017-03-20 (descending):
- No attempt: 10 | Failure (drone ship): 5 | Success (drone ship): 5 |
Controlled (ocean): 3 | Success (ground pad): 3 | Failure (parachute): 2 |
Uncontrolled (ocean): 2 | Precluded (drone ship): 1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky with stars and a view of the Earth's surface from space. The Earth's surface is mostly dark blue, with a thin layer of white clouds. A bright, glowing arc of city lights is visible along the horizon, indicating a coastal or urban area. The text "Section 3" is overlaid on the left side of the image.

Section 3

Launch Sites Proximities Analysis

Global Map: All Launch Sites

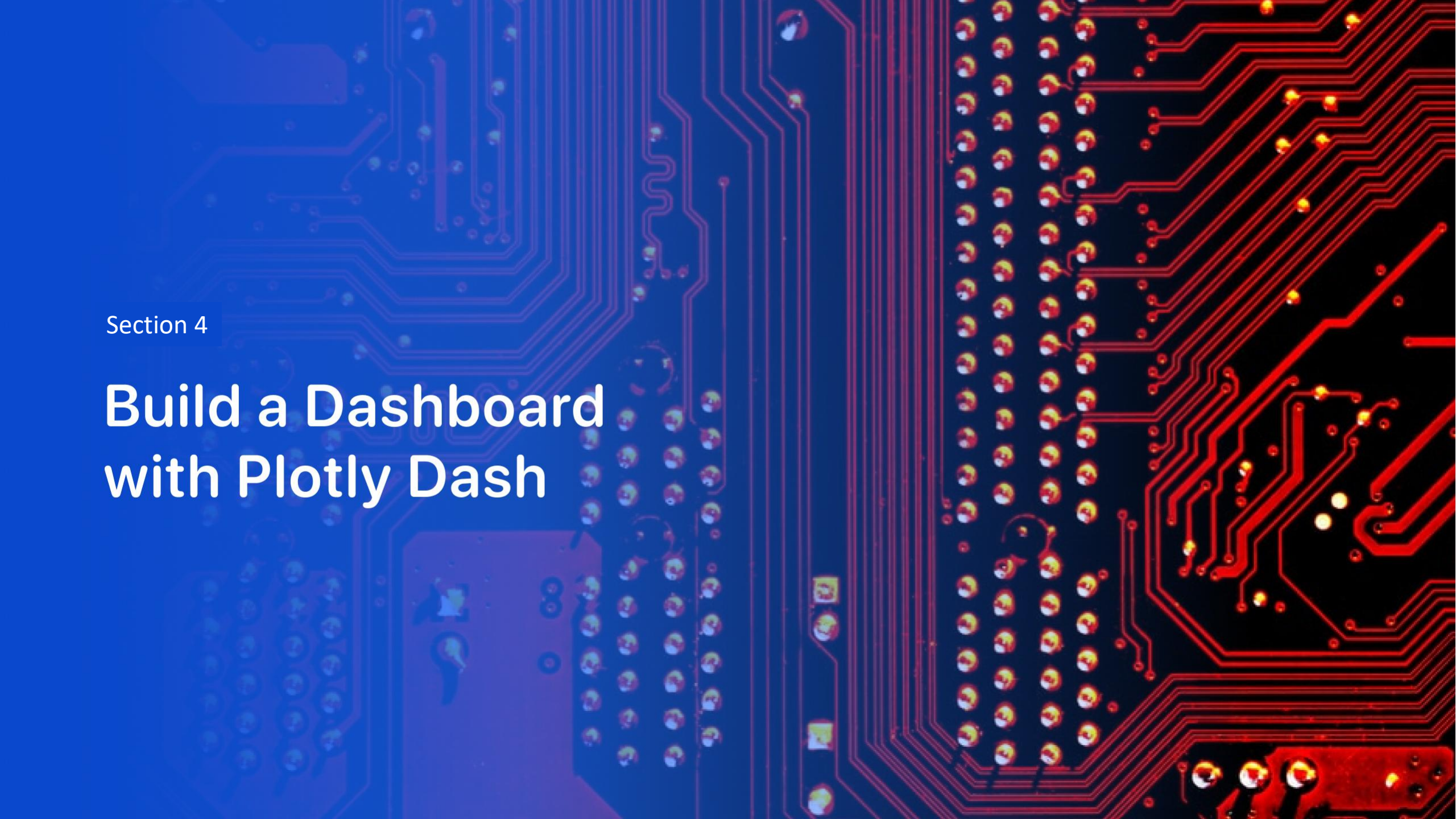
- All Launch Sites Location Markers on Global Map
- All launch sites near Equator and coastlines. Earth's equatorial rotation provides ~ 1670 km/h velocity boost for orbital launches.
- Sites: VAFB SLC-4E (California), CCAFS SLC-40 (Florida), KSC LC-39A (Florida). Equatorial proximity gives ~ 1670 km/h velocity boost.

Map: Success/Failure by Launch Site

- Colour-Labeled Launch Outcomes on Map
- Green = Success, Red = Failed. Folium Marker Cluster used to group nearby launches for readability.
- KSC LC-39A shows predominantly green markers (high success rate). CCAFS SLC-40 shows a mix of outcomes. Marker clusters expand on click to show individual launches.

Map: KSC LC-39A Proximity Analysis

- Distance from KSC LC-39A to Proximities
- KSC LC-39A distances: Railway = 15.23 km | Highway = 20.28 km | Coastline = 14.99 km | Closest city (Titusville) = 16.32 km.
- Failed rockets traveling at high speed could cover 15-20 km in seconds, posing potential danger. Coastal location ensures ocean-facing launches minimize risk to populated areas.



Section 4

Build a Dashboard with Plotly Dash

Pie Chart: Total Success Launches by Site

- Launch Success Count for All Sites
- Pie chart of total successful launches by site: KSC LC-39A leads at 41.2%.
Breakdown: CCAFS SLC-40 = 21.4%, VAFB SLC-4E = 23%, CCAFS LC-40 = 14.4%
- KSC LC-39A has the most successful launches out of all sites.

Pie Chart: KSC LC-39A Success Ratio

- Launch Site with Highest Launch Success Ratio
- KSC LC-39A piechart has 76.9% success rate — highest of all launch sites (10 successful, 3 failed).
- KSC LC-39A consistently outperforms other sites with 76.9% first-stage landing success rate.

Scatter Chart: Payload vs. Success by Booster

- Payload Mass vs. Launch Outcome for All Sites
- Scatter chart of Payload Mass vs. Success for different Booster Versions. Payloads between 2000-5500 kg show the highest success rate.
- Booster Version B5 (Block 5) has best success. Heavier payloads (>7000 kg) also show good success rates. Mid-range payloads 2000-5500 kg optimal.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

- All 4 models showed identical test scores (small sample). Decision Tree best overall: Jaccard=0.882, F1=0.938, Accuracy=0.911.
- On the entire dataset: Decision Tree is best — Jaccard=0.882, F1=0.938, Accuracy=0.911. SVM was second: Jaccard=0.845, F1=0.916, Accuracy=0.878.

Confusion Matrix

- Logistic Regression confusion matrix shows: $TN=3$, $FP=3$, $FN=0$, $TP=12$.
The major issue is false positives (predicting successful landing when it failed). All models can distinguish between classes but struggle with false positives.

Conclusions

- Decision Tree Model is the best algorithm for this dataset (Accuracy: 91.1%, F1: 0.9375).
- Launches with low payload mass show better results; KSC LC-39A has the highest success rate at 76.9%.
- Most launch sites are near the Equator and all are in close proximity to coastlines.
- The success rate of launches steadily increased from 2013 to 2020.
- Orbits ES-L1, GEO, HEO and SSO have 100% first-stage landing success rate.

Appendix

- Data sources: SpaceX REST API and Wikipedia Web Scraping. Tools used: Python, Pandas, NumPy, Matplotlib, Seaborn, Folium, Plotly Dash, Scikit-learn, BeautifulSoup, SQL (IBM Db2). GitHub: Data Collection API | Data Collection with Web Scraping | Data Wrangling | EDA with Data Visualization | EDA with SQL | Interactive Visual Analytics with Folium | SpaceX Dash App | Machine Learning Prediction. Special Thanks to Instructors, Coursera and IBM.

Thank you!

